



Fig. 3: GPS module

```
#include <avr/io.h>
#include <inttypes.h>
#include <avr/pgmspace.h>
#include "lcd2.h"
```

where '#include <avr/io.h>' is the AVR microcontroller header file for input/output (I/O) ports and '#include "lcd2.h"' is the header file for the LCD module.

The codes have been commented wherever necessary. For details of LCD interfacing and message display on the LCD, please refer the 'Moving Message Display On LCD' project published in EFY Dec. 2011 issue.

Data transfer between the GPS module and microcontroller through their respective transmit and receive pins is straightforward and therefore we have not discussed it here.

Compiling and programming the code. To work with the Atmel AVR microcontroller using 'C' programming language, you need two tools: AVR Studio and WinAVR. Both of these tools are freely available for download from the Internet.

AVR Studio is an integrated development environment that includes an editor, assembler, etc. It can be downloaded from the link <http://www.atmel.com/tools/atmelstudio.aspx?tab=overview>

WinAVR is a GCC-based compiler for AVR. It appears in AVR Studio as a plug-in. WinAVR also includes a program called Programmer's Notepad that can be used to edit and compile 'C' programs, independently of AVR Studio. WinAVR setup file is available

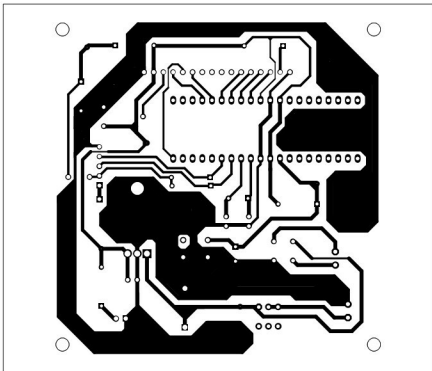


Fig. 4: An actual-size, single-side PCB for the standalone GPS receiver

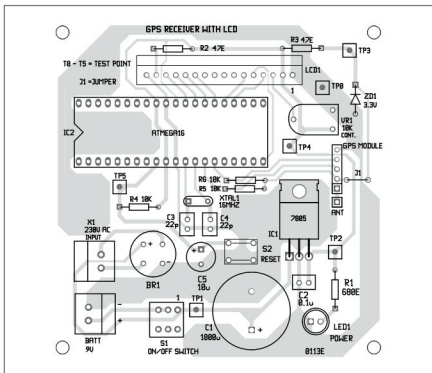


Fig. 5: Component layout for the PCB

at <http://winavr.sourceforge.net/>

Remember to install AVR Studio before WinAVR.

To compile the gpss.c code, first

launch AVR Studio from the desktop. You need to create a new project for the gpss.c code. Next, click 'Build→Rebuild All' to compile the